

Functions of Several Variables & Partial Derivatives

ANSWER KEY



Evaluating functions of two variables

- 1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.
 $f(2, -1) = (4)(-1) - 3(2)(-1) + 1 = -4 + 6 + 1 = 3$.
- 2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f .
 $f(3, 4) = \sqrt{9 + 16} = 5$. Domain: all of $\mathbb{R}^2$, since $x^2 + y^2 \geq 0$ always.

Computing partial derivatives

- 3 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y .
 $f_x = 4x^3y^3 + 2y$. $f_y = 3x^4y^2 + 2x - 5y^4$.
- 4 For $f(x, y) = e^{x^2y}$, find f_x and f_y .
 $f_x = 2xy e^{x^2y}$. $f_y = x^2 e^{x^2y}$.

Second-order and mixed partial derivatives

- 5 For $f(x, y) = x^3y^2$, find f_{xx} , f_{yy} , and f_{xy} .
 $f_x = 3x^2y^2$, $f_y = 2x^3y$. $f_{xx} = 6xy^2$. $f_{yy} = 2x^3$. $f_{xy} = 6x^2y$.

Clairaut's Theorem: equality of mixed partials

- 6 For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.
 $f_x = y\cos(xy) + 2xy$, so $f_{xy} = \cos(xy) - x\sin(xy) + 2x$. $f_y = x\cos(xy) + x^2$, so $f_{yx} = \cos(xy) - x\sin(xy) + 2x$. Equal, as expected.
- 7 State Clairaut's Theorem and the condition it requires.
If f_{xy} and f_{yx} are both continuous on an open region, then $f_{xy} = f_{yx}$ throughout that region.

Domains and level curves

- 8 Find the domain of $f(x, y) = \ln(x - y)$.
Requires $x - y > 0$, i.e. $x > y$. Domain: $\{(x, y) : x > y\}$.
- 9 Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.
 $x^2 + y^2 = 4$ is a circle of radius 2 centered at the origin.